

MANGU 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

231/3

BIOLOGY

PAPER 3 (PRACTICAL)

TIME: 1¾ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES:

- Write your **name** and **admission number** in the spaces provided.
- Sign and write **date** of examination in the spaces provided above
- Answer **all** the questions in section **in the spaces** provided after each question only
- You are required to spend the first 15 minutes of the 1 ¾ hours allowed for this paper, reading the whole paper carefully before starting to do this paper.

FOR EXAMINER'S USE ONLY:

QUESTIONS	MAXIMUM SCORE	CANDIDATE'S SCORE
1	16	
2	11	
3	13	
TOTAL	40	

1. You are provided with two test-tubes. Label them as A and B. You are required to prepare the contents of test tube A and B as follows. To test tube A, add 2 cm³ of solution R provided and test using the provided reagent. To test tube B, add 2 cm³ of solution T provided and test using the provided reagent.
(5 Marks)

(a)

Test tube	Procedure	Observation	Conclusion
A			
B			

(b) To the provided visking tubing tie one end with the provided string and add solution R. Tie the remaining end and immerse it in a solution of iodine solution in a beaker (50ml). After 5-10 minutes remove it from the beaker and observe.

(i) What was your observation of the contents of the visking tubing at the end of experiment?
(1 Mark)

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(ii) Account for your results in (i) above.
(4 Marks)

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(c) Explain **two** roles of physiological process under investigation in (c) above in living organisms.
(2 Marks)

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d) You are provided with specimen Q. Cut a longitudinal section specimen of specimen Q to make two parts. Add one drop of iodine solution on the surface of one of the exposed surfaces.

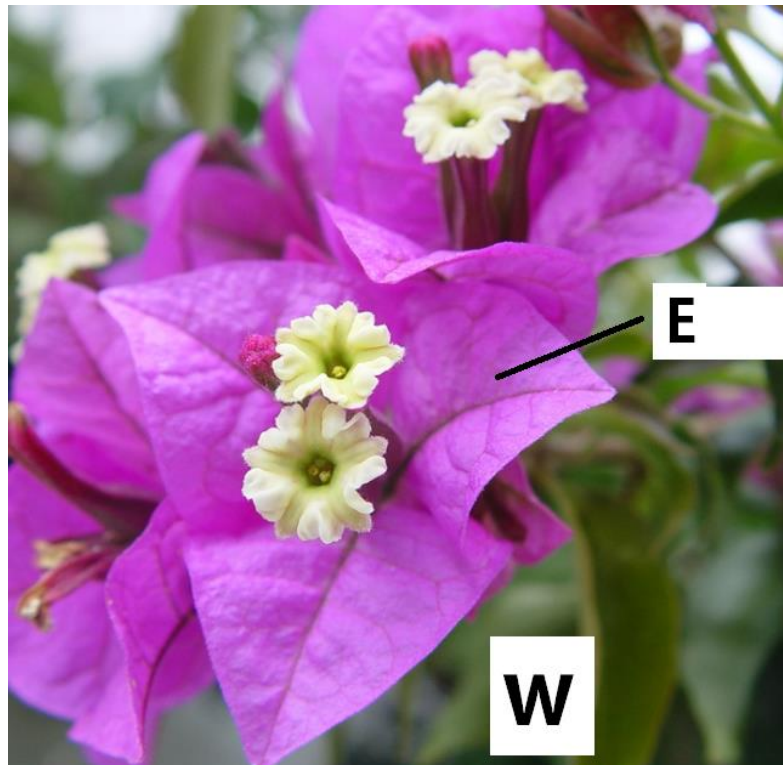
(i) State the observation
(1 Mark)

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(ii) Explain the observations made in d(i) above.
(3 Marks)

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2. The photographs below shows different types of flowers. Photograph V is a flower obtained from a Hibiscus plant while photograph W is a flower obtained from Bougainvillea plant.



(a) On the photograph, name the part labelled J and K
(2 Mark)

(b) (i) Name the agent of pollination for the flowers in the photograph (1 Mark)

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(ii) State **three** observations on the photographs that support the answer you have given in (b)(i) above
(3 Marks)

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(c) (i) Name the class of plant from which the photograph V was taken **(1 Mark)**

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(ii) Using observable features on the photograph, give the three reasons for your answer in c (i) above
(3 Marks)

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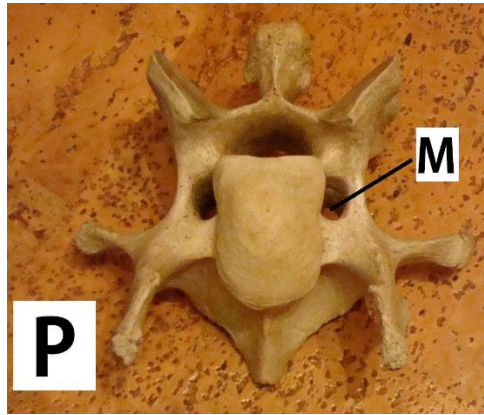
(iii) Name part labelled E on the photograph W
(1 Marks)

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3. The photographs below were taken from specimens obtained from mammalian skeleton. Evaluate them and use them for answering questions.



Y



(a) Identify the bone labelled X, Y, and T

(3 Marks)

X.....

Y.....

T

(b) State the function of part labelled M in bone P (1 Mark)

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(c) (i) Name the type of joint formed in the proximal end of bone T (1 Mark)

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(ii) State **two** characteristics of the joint named in c(i) above (2 Marks)

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(iii) On the image U, label **two** parts. (2 Marks)

(d) Discuss various adaptations of bone W to its functions (4 Marks)

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